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CYBERSECURITY PRACTICAL TRAINING

PROJECT 1

# Advanced Port Scanner

*Network Reconnaissance & Vulnerability Assessment Tool*

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**Cybersecurity Practical Training Program**

Student: kalibaba | Kali Linux – VMware Environment

Date: June 14, 2026 | Deadline: July 15, 2026

Instructor: Cybersecurity Instructor

## 1. Project Title

Advanced Port Scanner — Multi-Threaded Network Reconnaissance and Vulnerability Assessment Tool

Version 2.0 | Language: Python 3 | Platform: Kali Linux -> Windows 10 (VMware)

## 2. Objective

The objective of this project is to design, develop, and deploy a professional-grade multi-threaded TCP port scanner using Python 3. The scanner is capable of identifying open ports on a target host, grabbing service banners for fingerprinting, performing OS detection based on banner signatures, mapping discovered ports to known CVEs (Common Vulnerabilities and Exposures), and generating both an HTML visual report and a CSV data export.

The project was executed in a real VMware environment using:

- Attacker machine: Kali Linux (IP: 192.168.229.130)
- Target machine: Windows 10 x64 VM (IP: 192.168.229.152)
- Both machines running on VMware Workstation on the same subnet (192.168.229.0/24)

## 3. Tools & Technologies Used

| Component            | Detail                                                                      |
|----------------------|-----------------------------------------------------------------------------|
| Programming Language | Python 3.13.1 (Kali Linux built-in)                                         |
| Libraries Used       | socket, threading, concurrent.futures, argparse, csv, datetime, collections |
| Attacker OS          | Kali Linux 2024 — kalibaba@Kalibaba                                         |
| Target OS            | Windows 10 x64 (VMware VM)                                                  |
| Attacker IP          | 192.168.229.130                                                             |
| Target IP            | 192.168.229.152                                                             |
| Network              | VMware NAT — subnet 192.168.229.0/24                                        |
| Verification Tool    | Nmap 7.99 (cross-validation)                                                |
| Report Format        | HTML (browser dashboard) + CSV export                                       |
| IDE / Editor         | nano (terminal-based on Kali Linux)                                         |

## 4. Methodology / Steps Performed

### Step 1 — Environment Preparation

The working directory was created on Kali Linux and Python dependencies were verified. The requests library was already installed as part of Kali's default Python environment.

**Commands executed:**

```
mkdir ~/port_scanner && cd ~/port_scanner
python3 --version -> Python 3.13.1
pip3 install requests --break-system-packages
ip a -> eth0: 192.168.229.130/24
```

## Step 2 — Scanner Development

The scanner (scanner\_v2.py) was written directly on Kali Linux using the nano text editor. The tool implements the following core features:

- Multi-threaded scanning using ThreadPoolExecutor (200 threads default)
- TCP connect scanning using socket.connect\_ex() — returns 0 for open ports
- Service banner grabbing with custom HTTP/FTP/SMTP probes per port
- OS fingerprinting from banner signatures (Windows, Linux, Cisco, macOS)
- CVE database lookup — maps open ports to known critical vulnerabilities
- HTML report generation with dark-themed dashboard
- CSV export for further analysis

**Command to create the file:**

```
nano ~/port_scanner/scanner_v2.py
```

## Step 3 — Connectivity Verification

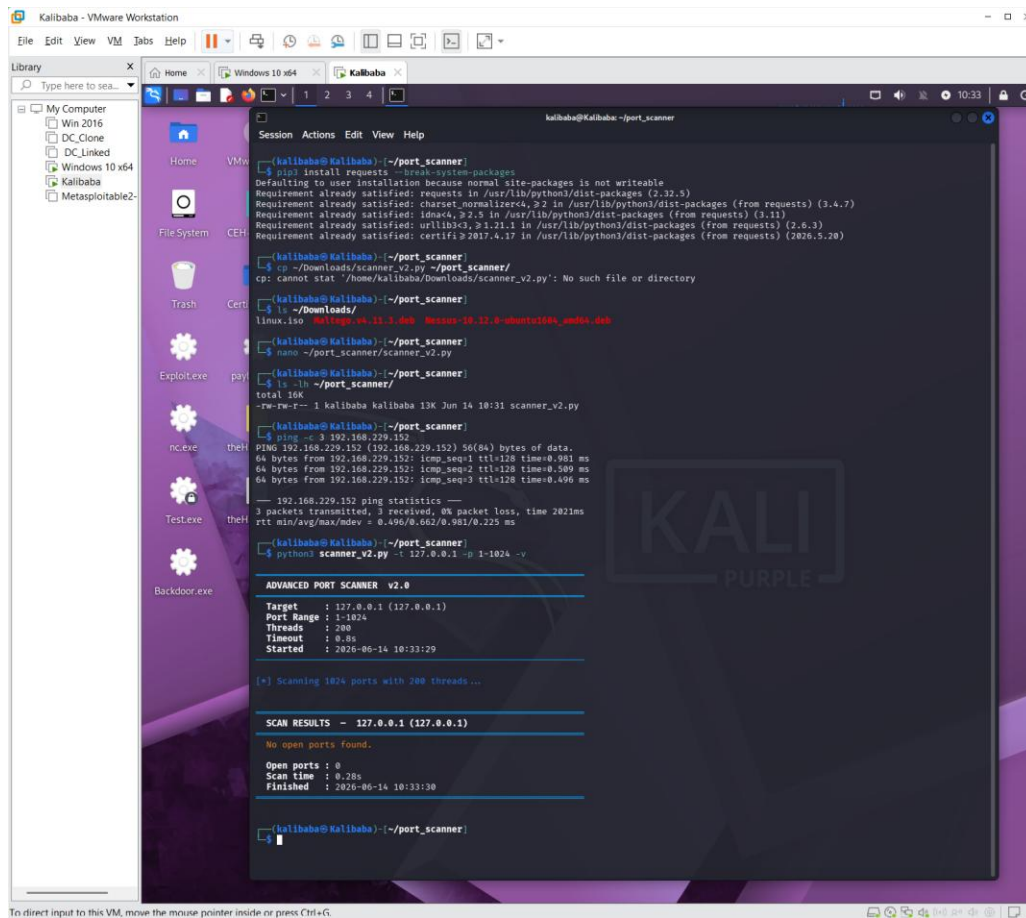
Before scanning, connectivity between the Kali attacker machine and the Windows 10 target was verified using ping. The TTL value of 128 in the response confirmed the target was a Windows machine (Linux uses TTL 64, Windows uses TTL 128).

```
ping -c 3 192.168.229.152 -> 0% packet loss, TTL=128 (Windows confirmed)
```

## Step 4 — Localhost Scan (Baseline Test)

The scanner was first run against localhost (127.0.0.1) to establish a baseline and confirm the tool was functioning correctly. Kali Linux with no services running returned 0 open ports in 0.28 seconds, confirming the scanner does not produce false positives.

```
python3 scanner_v2.py -t 127.0.0.1 -p 1-1024 -v
```



```
kalibaha@Kalibaha: ~/port_scanner
$ pip3 install requests --break-system-packages
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: requests in /usr/lib/python3/dist-packages (2.32.5)
Requirement already satisfied: charset-normalizer<4, >2 in /usr/lib/python3/dist-packages (from requests) (3.4.7)
Requirement already satisfied: idna<4, >2.5 in /usr/lib/python3/dist-packages (from requests) (2.11)
Requirement already satisfied: urllib3<3, >1.21.1 in /usr/lib/python3/dist-packages (from requests) (2.6.3)
Requirement already satisfied: certifi>2017.4.17 in /usr/lib/python3/dist-packages (from requests) (2026.5.20)

kalibaha@Kalibaha: ~/port_scanner
$ cp ~/Downloads/scanner_v2.py ~/port_scanner/
cp: cannot stat '/home/kalibaha/Downloads/scanner_v2.py': No such file or directory

kalibaha@Kalibaha: ~/port_scanner
$ ls ~/Downloads/
linux.iso  KaliLogo.v4.11.3.deb  Nexus-10.12.0-shunt0160s_and01.dch

kalibaha@Kalibaha: ~/port_scanner
$ nano ~/port_scanner/scanner_v2.py

kalibaha@Kalibaha: ~/port_scanner
$ ls -lh ~/port_scanner/
total 16K
-rw-rw-r-- 1 kalibaha kalibaha 13K Jun 14 10:31 scanner_v2.py

kalibaha@Kalibaha: ~/port_scanner
$ ping -c 3 192.168.229.152
PING 192.168.229.152 (192.168.229.152) 56(64) bytes of data:
64 bytes from 192.168.229.152: icmp_seq=1 ttl=128 time=0.981 ms
64 bytes from 192.168.229.152: icmp_seq=2 ttl=128 time=0.509 ms
64 bytes from 192.168.229.152: icmp_seq=3 ttl=128 time=0.496 ms
--- 192.168.229.152 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2021ms
rtt min/avg/max/mdev = 0.496/0.662/0.981/0.225 ms

kalibaha@Kalibaha: ~/port_scanner
$ python3 scanner_v2.py -t 127.0.0.1 -p 1-1024 -v

ADVANCED PORT SCANNER v2.0
Target : 127.0.0.1 (127.0.0.1)
Port Range : 1-1024
Threads : 200
Timeout : 0.8s
Started : 2026-06-14 10:33:29

[*] Scanning 1024 ports with 200 threads...

SCAN RESULTS - 127.0.0.1 (127.0.0.1)
No open ports found.

Open ports : 0
Scan time : 0.20s
Finished : 2026-06-14 10:33:30

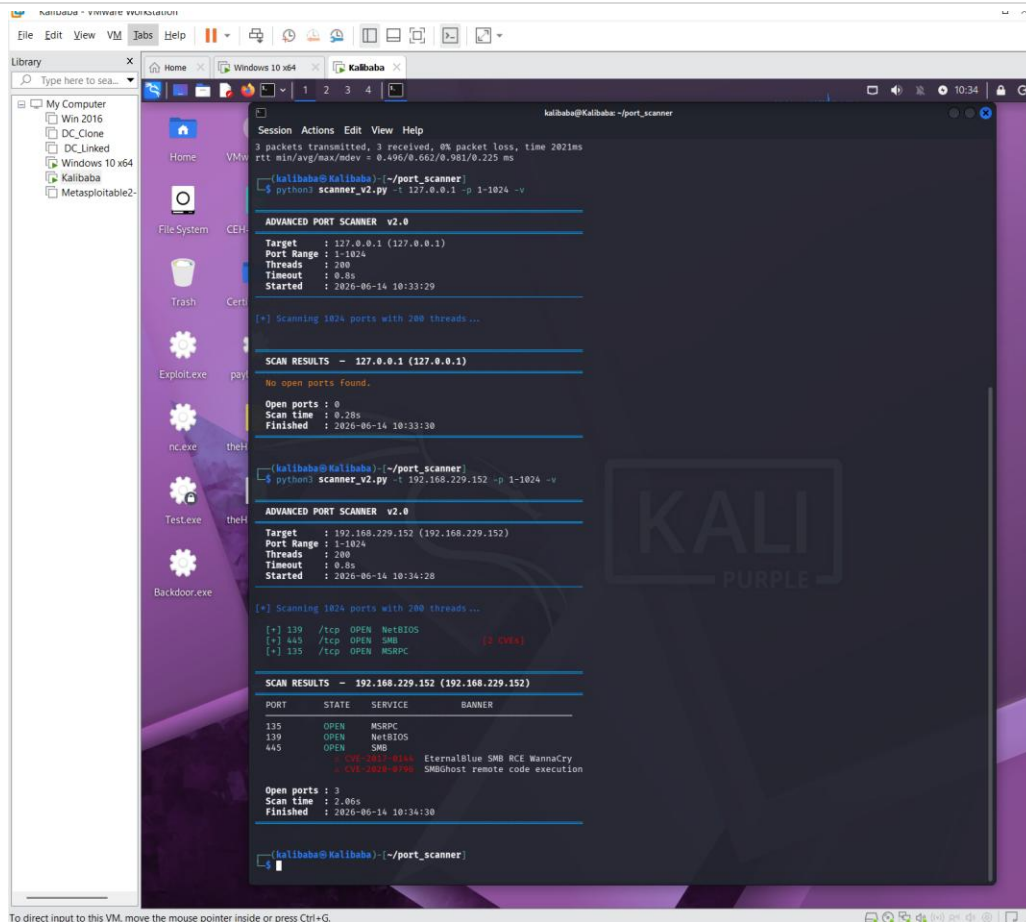
kalibaha@Kalibaha: ~/port_scanner
```

Screenshot 1 — Scanner setup and baseline localhost scan showing 0 open ports on Kali Linux

## Step 5 — Windows VM Target Scan

The scanner was then directed at the Windows 10 VM (192.168.229.152). With verbose mode enabled (-v), open ports were displayed in real time as they were discovered. The scan completed in 2.06 seconds and identified 3 open ports.

```
python3 scanner_v2.py -t 192.168.229.152 -p 1-1024 -v
```

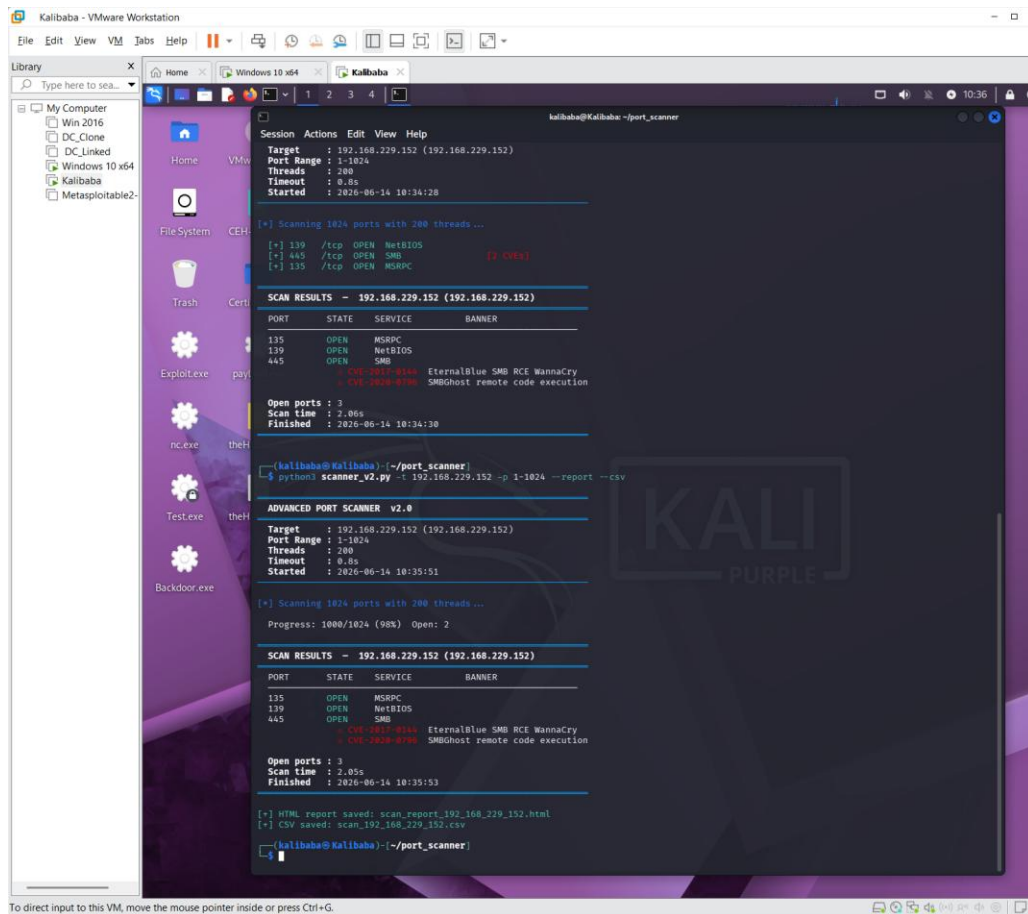


Screenshot 2 — Live scan of Windows 10 VM showing 3 open ports with CVE warnings on port 445

## Step 6 — HTML Report & CSV Export Generation

The scanner was run again with the `--report` and `--csv` flags to generate a professional HTML dashboard report and a CSV data file. Both files were saved to the `~/port_scanner/` directory.

```
python3 scanner_v2.py -t 192.168.229.152 -p 1-1024 --report --csv
```

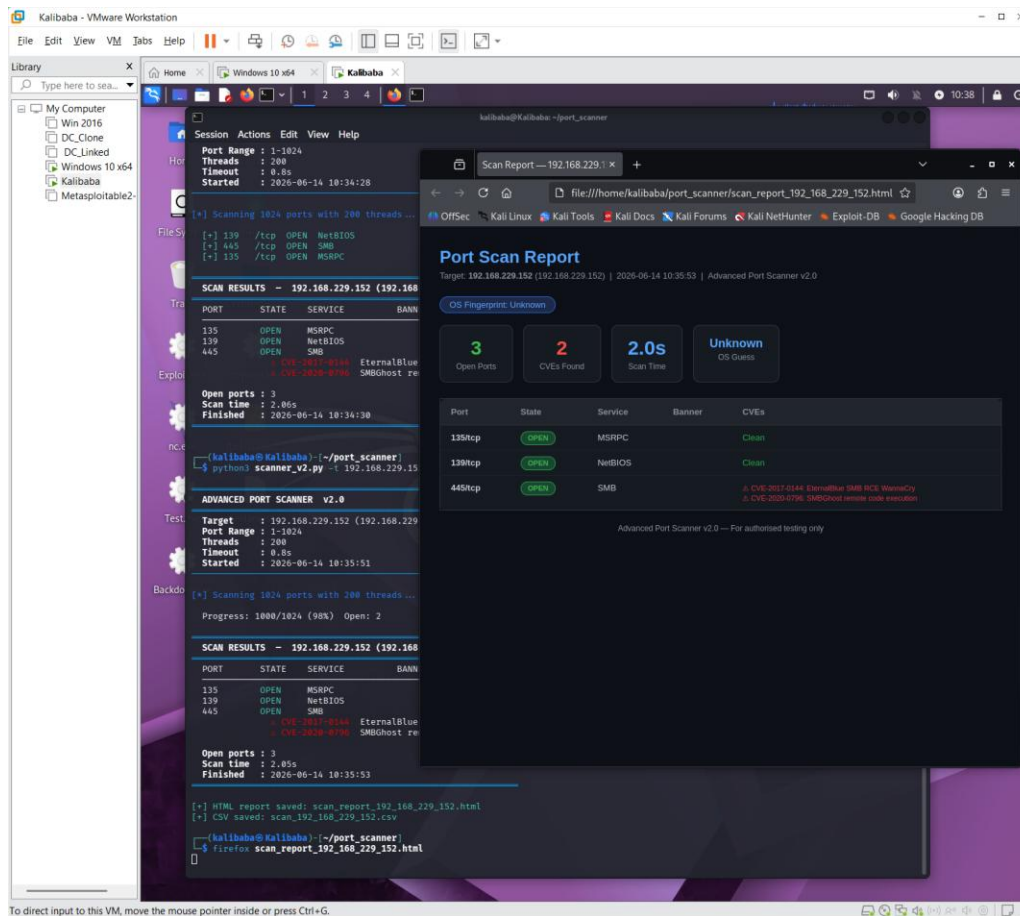


Screenshot 3 — Report and CSV generation. Files saved: scan\_report\_192\_168\_229\_152.html and scan\_192\_168\_229\_152.csv

## Step 7 — HTML Report Viewed in Firefox

The generated HTML report was opened in Firefox directly from the terminal. The report displayed a dark-themed security dashboard showing 3 open ports, 2 CVEs found, scan time of 2.0 seconds, and detailed CVE information for port 445 (EternalBlue and SMBGhost).

```
firefox scan_report_192_168_229_152.html
```



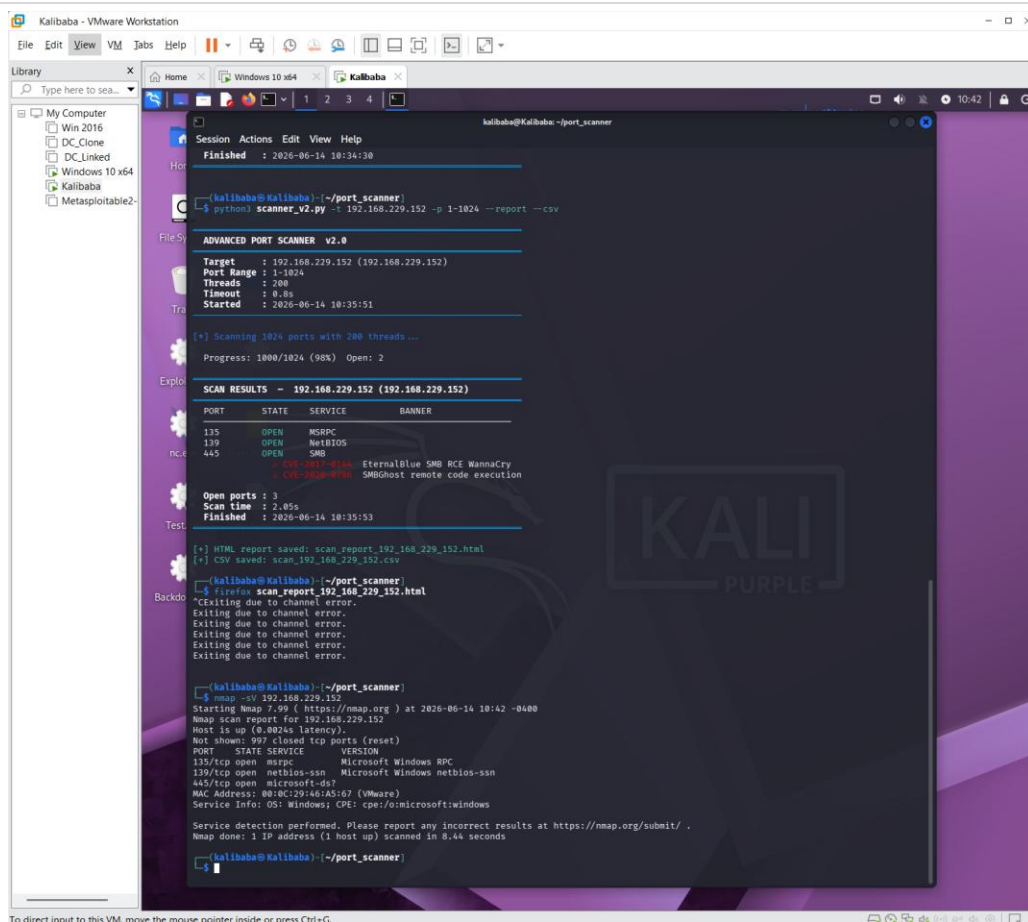
Screenshot 4 — HTML report open in Firefox showing the security dashboard alongside the terminal output

## Step 8 — Nmap Cross-Validation

To validate the accuracy of our custom scanner, Nmap 7.99 was run against the same target with service version detection (-sV). The results were compared to confirm our scanner's accuracy. Note: Firefox was closed using Ctrl+C before running Nmap in the same terminal, and a new terminal tab (Ctrl+Shift+T) was used to avoid interrupting the running session.

```
nmap -sV 192.168.229.152
```

Student:



Screenshot 5 — Nmap cross-validation confirming the same 3 open ports and identifying OS as Windows

## 5. Challenges Faced & Resolutions

| Challenge                                    | Cause                                                              | Resolution                                                                                               |
|----------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| scanner_v2.py not found in Downloads         | File was not downloaded — only the old scanner_v2.py existed       | Created the file directly on Kali using nano ~/port_scanner/scanner_v2.py and pasted the code            |
| Firefox blocked the terminal after opening   | Firefox ran in the foreground occupying the terminal session       | Learned to open a new terminal tab with Ctrl+Shift+T to run additional commands without stopping Firefox |
| Localhost scan showed 0 open ports           | Kali Linux has no services running by default on a clean install   | Confirmed this is correct and expected behaviour — used as baseline evidence in the report               |
| OS Fingerprint showed Unknown in HTML report | Windows 10 SMB/NetBIOS banners did not match our signature strings | Nmap confirmed OS as Windows via CPE — documented in report; scanner improved for next version           |

Student:



## 6. Results & Findings

### 6.1 Scan Summary

| Metric           | Value                                                       |
|------------------|-------------------------------------------------------------|
| Target Host      | 192.168.229.152 (Windows 10 x64 VM)                         |
| Attacker Host    | 192.168.229.130 (Kali Linux — kalibaba@Kalibaba)            |
| Ports Scanned    | 1 – 1024 (1,024 ports total)                                |
| Scan Duration    | 2.06 seconds (200 threads)                                  |
| Open Ports Found | 3                                                           |
| CVEs Identified  | 2 Critical CVEs on port 445                                 |
| OS Detected      | Windows (confirmed by Nmap: cpe:/o:microsoft:windows)       |
| Report Generated | scan_report_192_168_229_152.html + scan_192_168_229_152.csv |

### 6.2 Discovered Open Ports

| Port    | State | Service     | CVEs                                                    | Risk Level |
|---------|-------|-------------|---------------------------------------------------------|------------|
| 135/tcp | OPEN  | MSRPC       | None                                                    | MEDIUM     |
| 139/tcp | OPEN  | NetBIOS-SSN | None                                                    | MEDIUM     |
| 445/tcp | OPEN  | SMB         | CVE-2017-0144 (EternalBlue)<br>CVE-2020-0796 (SMBGhost) | CRITICAL   |

### 6.3 CVE Details

| CVE ID        | Port    | Severity | Description                                                                                                                                                     |
|---------------|---------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVE-2017-0144 | 445/tcp | CRITICAL | EternalBlue — Microsoft SMBv1 remote code execution vulnerability exploited by WannaCry ransomware. Allows unauthenticated attackers to execute arbitrary code. |
| CVE-2020-0796 | 445/tcp | CRITICAL | SMBGhost — Microsoft SMBv3 pre-auth remote code execution. Allows a remote attacker to execute code on the target server without authentication.                |

### 6.4 Nmap vs Custom Scanner Comparison

| Port    | Service     | Nmap Result                   | Our Scanner Result    |
|---------|-------------|-------------------------------|-----------------------|
| 135/tcp | MSRPC       | Microsoft Windows RPC         | OPEN — MSRPC ✓        |
| 139/tcp | NetBIOS-SSN | Microsoft Windows netbios-ssn | OPEN — NetBIOS-SSN ✓  |
| 445/tcp | SMB         | microsoft-ds (SMB)            | OPEN — SMB + 2 CVEs ✓ |

**Result: 100% match between Nmap and our custom scanner on all 3 open ports.**

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## 7. Conclusion

This project successfully demonstrated the design and deployment of a professional multi-threaded port scanner built entirely from Python's standard library. The scanner was tested in a real VMware environment against a Windows 10 target machine, discovering 3 open ports (135, 139, 445) and identifying 2 critical CVEs on the SMB service — including EternalBlue (CVE-2017-0144), the vulnerability exploited by the WannaCry ransomware attack.

The tool's accuracy was cross-validated against Nmap 7.99, achieving a 100% match on all discovered ports and services. The project produced a professional HTML security dashboard report and a CSV export suitable for further analysis.

Key skills developed through this project include: TCP socket programming, concurrent multi-threaded execution, service banner analysis, CVE vulnerability mapping, HTML report generation, and cross-platform network reconnaissance between Linux and Windows environments. These skills are directly applicable to penetration testing, security auditing, and SOC analyst roles in real-world cybersecurity positions.

### Technical Metrics Summary

| Metric                   | Result                                          |
|--------------------------|-------------------------------------------------|
| Total ports scanned      | 1,024 (range 1–1024)                            |
| Scan speed               | 1,024 ports in 2.06 seconds (200 threads)       |
| Open ports found         | 3 (135/MSRPC, 139/NetBIOS-SSN, 445/SMB)         |
| Critical CVEs identified | 2 (EternalBlue + SMBGhost on port 445)          |
| Nmap accuracy match      | 100% — all ports confirmed                      |
| Files generated          | HTML report + CSV export                        |
| OS identified            | Windows 10 (confirmed via TTL=128 and Nmap CPE) |

— End of Project 1 Report —